CS5487 . 08 _ Discriminative Learning
Linear Classifiers.

Generative Models:
1) Learn CCD from training data $p(x|y)$
2) use BDT to get decision rule:
calculate $p(y|x)$ $g(x) = \arg\max p(y|x)$

Discriminative Solution:
Solve for the decision rule directly:
$p(y|x)$ or $g(x)$

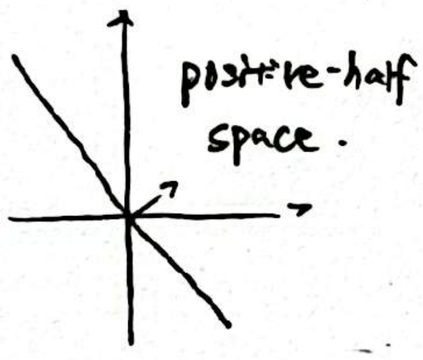
Linear Classifier:
binary class $y \in \{-1, 1\}$.
input : $x \in \mathbb{R}$

$f(x) = w^T x$ parameter $w \in \mathbb{R}^d$

$\Downarrow$

$f(x) = \tilde{w}^T \tilde{x} = w^T x + b$

$\tilde{w} = \begin{bmatrix} w \\ b \end{bmatrix}$ $\tilde{x} = \begin{bmatrix} x \\ 1 \end{bmatrix}$



Decision rule:

$y^* = \text{sign}(w^T x)$ $\begin{cases} +1 & w^T x > 0 \\ -1 & w^T x < 0 \end{cases}$

Training set: $D = \{(x_i, y_i)\}_{i=1}^n$

Note: given a w, $y_i w^T x_i > 0$
$(x_i y_i)$ correctly classified.
$y_i w^T x_i < 0$ misclassified.

Idea: 0-1 loss function

$w^* = \arg\min_w \sum_{i=1}^n \begin{cases} 0 & y_i w^T x_i \geq 0 \\ 1 & y_i w^T x_i < 0 \end{cases} = \#\text{mistakes}$

To difficult to optimize:
gradient = {0, undifined}

Label regression: (Least squares Classification)
Problem: $\hat{w} = \arg\min_w \sum_{i=1}^n (w^T x_i - y_i)^2$

$\hat{w} = (XX^T)^{-1} Xy$

Not robust to "outliers"
FLD is related to LSC

perception:
perceperon criterion: only look at misclassified samples

$E(w) = \sum_{i \in M} -y_i w^T x_i$ $M = \{ i \mid y_i w^T x_i < 0 \}$

$E(w) = 0$: training set is ~~are~~ correctly classified

Perceptron Algorithm:

$w^* = \arg\min \sum_{i \in M} -y_i w^T x_i$

$w_{t+1} = w_t + \eta y_i x_i$ for $i \in M$.

How to see η : it doesn't mater.

- Rosenblatt:
SGD converges on $\left(\frac{R^2}{\gamma^2}\right)$ iterations if the data is linearly separatable.

$R = \max \|x_i\|$

γ = "margin"

- will not converge if the data is not linearly separatable.
- many possible solutions. (depends on initialization)

Logistic Regression: (Probabilitistic approach)
Consider $y \in \{0, 1\}$
: when CCDs are Gaussian $p(x|y) = N(\mu_y, \Sigma_y)$
the posterior $p(y|x)$ is a sigmoid function

$p(y=1|x) = \frac{1}{1+e^{-(w^T x + b)}}$

$\Rightarrow f(x) = w^T x$. $p(y=1|x) = \frac{1}{1+e^{-w^T x}}$

$= \sigma(w^T x) = \pi$

$\sigma(z) = \frac{1}{1+e^{-z}}$

Decision Rule: $\hat{y} = \begin{cases} 1 & p(y=1|x) > \frac{1}{2} \\ 0 \end{cases}$

$= \begin{cases} 1 & w^T x > 0 \\ 0 & \text{otherwise} \end{cases}$.

BDR - Gaussian: $O(d^2)$
Logistic Regression $O(d)$ ✓

Learning:

let $\pi_i = \sigma(w^T x_i)$

Bernoulli likelihood $P(y_i | x_i, w) = \pi_i^{y_i} (1-\pi_i)^{1-y_i}$

Data log likelihood:

$$\ell(w) = \sum_{i=1}^{n} \log P(y_i | x_i, w) = \sum_{i=1}^{n} y_i \log \pi_i + (1-y_i) \log(1-\pi_i)$$

• MLE: $w^* = \arg\max_{w} \ell(w)$

$$w^* = \arg\min_{w} \sum_{i=1}^{n} \underbrace{-y_i \log \pi_i - (1-y_i) \log(1-\pi_i)}_{\text{binary cross entropy Loss}}$$